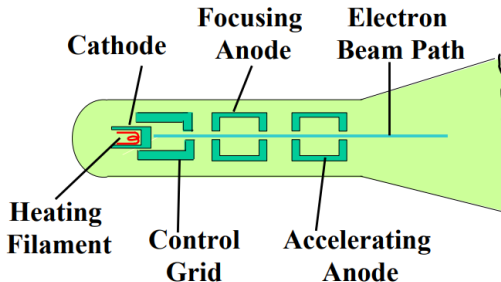


Computer Graphics

Display devices and types of display

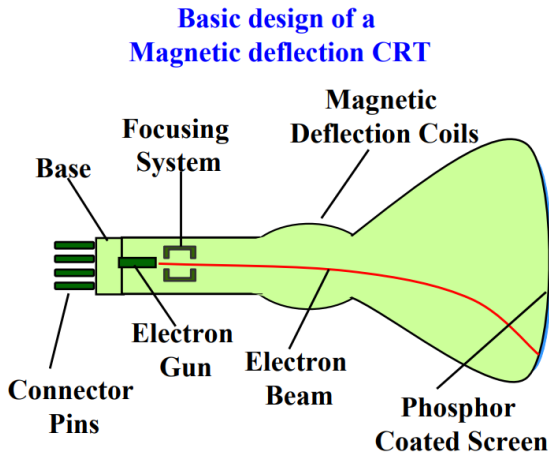
Course Instructor:
Dr. Suman Kumar Maji

Operation of an electron gun with an accelerating anode



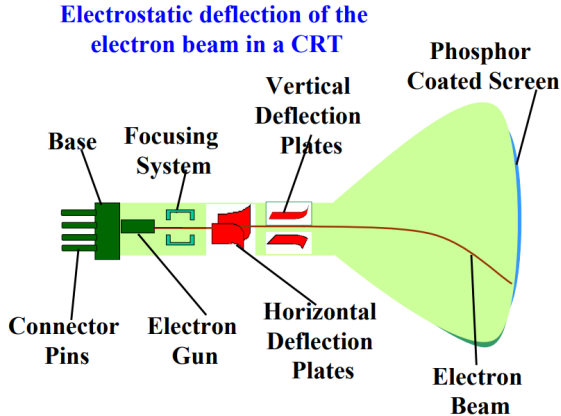
Slide courtesy: Prof. Sukhendu Das IIT Madras

Basic Design of a magnetic deflection CRT



Slide courtesy: Prof. Sukhendu Das IIT Madras

Electrostatic deflection of the electron beam in a CRT



Slide courtesy: Prof. Sukhendu Das IIT Madras

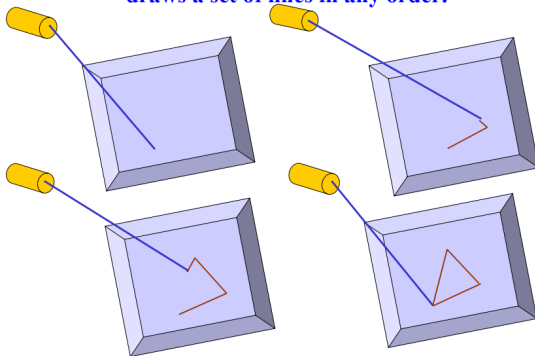
Types of display

Display can be classified into two types:

- Random scan display (or calligraphic display)
- Raster scan display

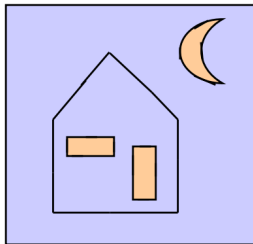
Calligraphic or Random Scan display system

**Random-scan display system
draws a set of lines in any order.**

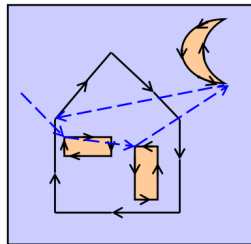


Slide courtesy: Prof. Sukhendu Das IIT Madras

Ideal line drawing v/s Vector scan



(a) Ideal line drawing



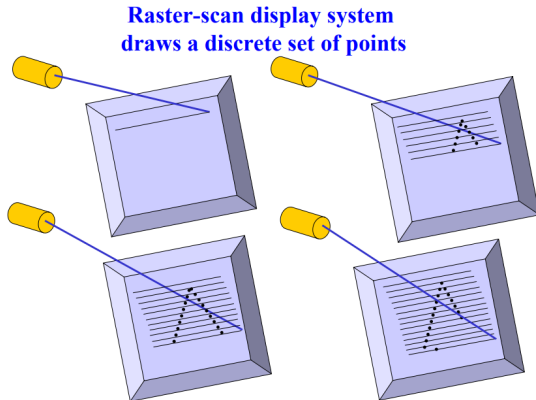
(b) Vector scan

Slide courtesy: Prof. Sukhendu Das IIT Madras

Raster scan display system

- Entire screen is a matrix of pixels
- Each pixel brightness can be controlled
- Refresh buffer can be visualized as a set of horizontal raster lines or a row of individual pixels
- Line cannot be drawn directly from one point to another
- Each point is an addressable point in screen and memory
- This causes the effect of 'aliasing', 'jaggies' or 'staircase' effect
- Refresh/Frame buffer is also called Bit-plane

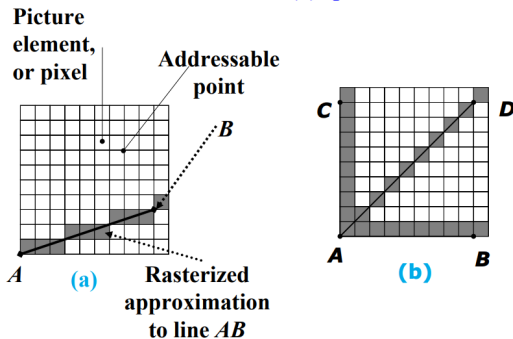
Non-interlacing display



Slide courtesy: Prof. Sukhendu Das IIT Madras

Non-interlacing display

Rasterization: (a) General line ;
(b) special cases



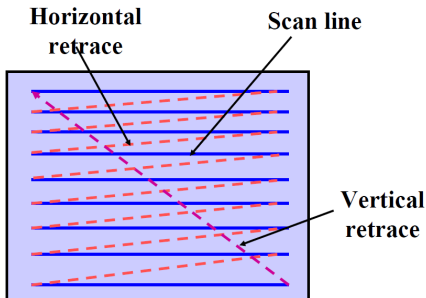
Slide courtesy: Prof. Sukhendu Das IIT Madras

Non-interlacing display

- Raster is stored as a matrix of pixels representing the entire screen area
- The entire image is scanned out sequentially by the video controller (one raster line at a time)
- The raster lines are scanned from top to bottom and then back to the top
- The intensity of the beam decides the brightness of the pixel
- At least one memory bit for each pixel (called bitplane)

Non-interlacing display

Raster Scan

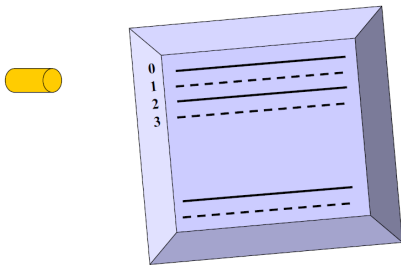


Slide courtesy: Prof. Sukhendu Das IIT Madras

Interlacing display

Interlacing scan lines on a raster scan display;

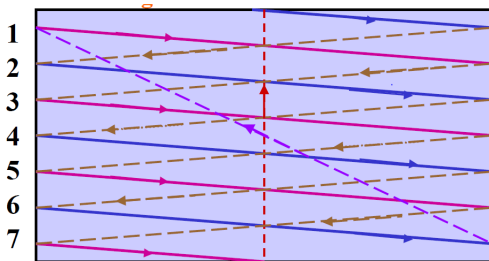
First, all points on the even-numbered (solid) scan lines are displayed;
then all points along the odd-numbered (dashed) lines are displayed



Slide courtesy: Prof. Sukhendu Das IIT Madras

Schematic of a 7-line interlaced scan line pattern

The odd field begins with line 1. The horizontal retrace is shown dashed. The odd field vertical retrace starts at the bottom center. The even field vertical retrace starts at the bottom right.



Slide courtesy: Prof. Sukhendu Das IIT Madras

Refresh rate, video basics and scan conversion example

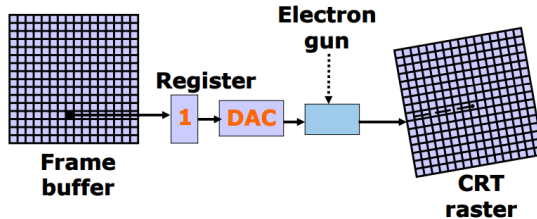
A typical example:

If one uses a 512x512 element raster display, then 2^{18} bits are necessary in a single bit plane. Memory size required: 32 KB.

A DAC (digital-to-analog converter) is used to convert the bit value (0, 1) to analog signals for refreshing the screen. Memory size required for N-bit plane gray level frame buffers:

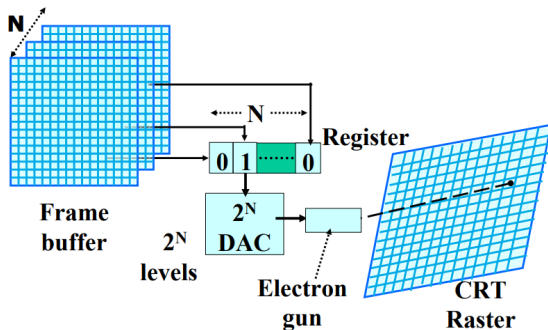
N	Size in KB
3	96
8	256
24	768

A single bit-plane black & white frame buffer raster CRT graphics device.



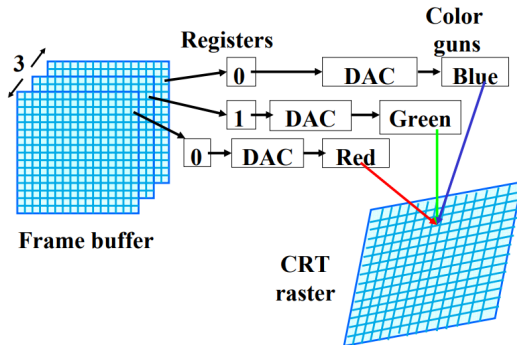
Slide courtesy: Prof. Sukhendu Das IIT Madras

An N-bit plane gray level frame buffer



Slide courtesy: Prof. Sukhendu Das IIT Madras

Simple color frame buffer



Slide courtesy: Prof. Sukhendu Das IIT Madras

N-bit plane gray level Frame buffer

In case of one-bit for each color frame buffer, we get 8 colors as:

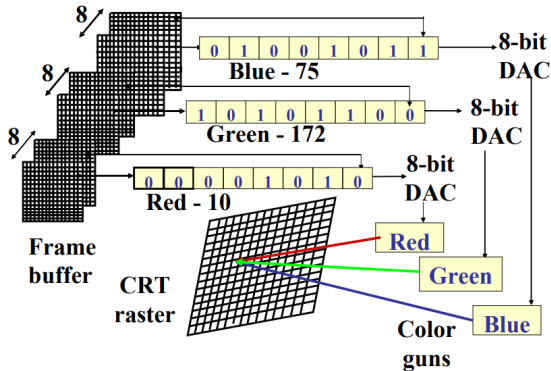
COLOR	RED	GREEN	BLUE
BLACK	0	0	0
BLUE	0	0	1
GREEN	0	1	0
CYAN	0	1	1
RED	1	0	0
MAGENTA	1	0	1
YELLOW	1	1	0
WHITE	1	1	1

Slide courtesy: Prof. Sukhendu Das IIT Madras

N-bit plane gray level Frame buffer

- Typically 8-bit planes per color is used, which gives a 24-bit plane frame buffer
- Each group of bit-planes drives an 8-bit DAC
- Each group generates 256 shades of intensities of red, green or blue
- Hence we obtain $2^{24} = 16,777,216$ possible colors.
- This is called a **FULL COLOR FRAME BUFFER**

A 24-bit-plane color frame buffer



Slide courtesy: Prof. Sukhendu Das IIT Madras